

Theoretical analysis of high harmonic generation in solids: supplementary material

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Derivation of two-band equations

Our analysis of the interaction of strong lasers with solids is based on a three dimensional, two-band model solid coupled to a laser field $\mathbf{F}(t) = \hat{\mathbf{x}}F_0f(t)$. The laser is characterized by field amplitude $f(t)$, peak field strength in the material F_0 , and polarization along $\hat{\mathbf{x}}$. The dipole approximation is used, as the laser wavevector ($\mathbf{k} \parallel \hat{\mathbf{z}}$) is much smaller than the reciprocal lattice vectors. Atomic units are used throughout, if not explicitly noted otherwise.

We will first derive the two-band equations starting from the Schrödinger equations [1] and from there will derive the two-band density matrix equations [2]. The effect of the laser field can be accounted for in two ways. In the length gauge the time dependent Hamiltonian is given by $H(t) = H_0 - \mathbf{x} \cdot \mathbf{F}(t)$, where $H_0 = T + U$ is the unperturbed Hamiltonian with $T = -1/2\nabla^2$ the kinetic energy operator, $\hat{\mathbf{p}} = -i\nabla$ the momentum operator, and $U(\mathbf{x})$ the periodic potential of the lattice. In the velocity gauge $\hat{\mathbf{p}}$ is replaced by $\hat{\mathbf{p}} - \mathbf{A}$; the vector potential $\mathbf{A}$ is defined by $\mathbf{F} = -d\mathbf{A}/dt$.

The field free Hamiltonian H_0 has Bloch eigenstates $\Phi_{\mathbf{m},\mathbf{k}}(\mathbf{x}) = u_{\mathbf{m},\mathbf{k}}(\mathbf{x}) \exp(i\mathbf{k} \cdot \mathbf{x})$ with energies $E_{\mathbf{m},\mathbf{k}} = E_{\mathbf{m}}(\mathbf{k})$ in band $\mathbf{m}$ with crystal momentum $\mathbf{k}$; $u_{\mathbf{m},\mathbf{k}}$ is the periodic part of the Bloch function. Our analysis is restricted to a two-band model with valence ($\mathbf{m} = v$) and conduction ($\mathbf{m} = c$) band.

In the presence of the laser field the wavefunction becomes time dependent. In the length gauge it is represented as

$$\Psi(\mathbf{x}, t) = \sum_{\mathbf{m}=c,v} \int_{\text{BZ}} a_{\mathbf{m}}(\mathbf{k}, t) \Phi_{\mathbf{m},\mathbf{k}}(\mathbf{x}) d^3\mathbf{k} \quad (1)$$

with probability amplitudes $a_{\mathbf{m}}(\mathbf{k}, t)$. The time dependent wavefunctions obtained from the length gauge (Ψ) and velocity gauge (Ψ_v) Hamiltonian differ by a phase factor, $\Psi = \Psi_v \exp(-i\mathbf{A} \cdot \mathbf{r})$. The phase factor leaves expectation values of operators unchanged, and therewith the same results are obtained for laser solid interaction processes in the length and velocity gauge. We will show the derivation for the length gauge in the following. The derivation in the velocity gauge was done in [1] and yields the same equations for the probability amplitudes. However the Bloch eigenstates are multiplied with the above phase factor.

To obtain the equations of motion for the probability amplitudes $a_{\mathbf{m}}(\mathbf{k}, t)$, ansatz (1) is substituted into the

TDSE for the length gauge Hamiltonian $H(t)$, and the Bloch eigenstates are integrated out which yields

$$\dot{a}_{\mathbf{m}} = (-iE_{\mathbf{m}}(\mathbf{k}) + \mathbf{F}(t) \cdot \nabla_{\mathbf{k}}) a_{\mathbf{m}} + i\mathbf{F}(t) \cdot \sum_{\mathbf{m}' \neq \mathbf{m}} \mathbf{d}_{\mathbf{m}\mathbf{m}'}(\mathbf{k}) a_{\mathbf{m}'} \quad (2)$$

with $\mathbf{m} = v, c$, $\mathbf{m}' = v, c$, and with

$$\mathbf{d}_{\mathbf{m}\mathbf{m}'}(\mathbf{k}) = i \int d^3\mathbf{x} u_{\mathbf{m},\mathbf{k}}^*(\mathbf{x}) \nabla_{\mathbf{k}} u_{\mathbf{m}',\mathbf{k}}(\mathbf{x}) \quad (3)$$

the transition dipole moment between valence and conduction bands. In the derivation we used the relation $\langle \Phi_{\mathbf{m},\mathbf{k}} | \mathbf{x} | \Phi_{\mathbf{m}',\mathbf{k}'} \rangle = i\partial_{\mathbf{k}} u_{\mathbf{m}} + \mathbf{d}_{\mathbf{m}\mathbf{m}'}(\mathbf{k})$ [3], which splits the transition dipole into an intraband and interband contribution, respectively. For a direct numerical calculation of the dipole moment $d_{\mathbf{m}\mathbf{m}'}$, see Ref. [4]. Below we show how $d_{\mathbf{m}\mathbf{m}'}$ can be evaluated from the momentum matrix element, which is tabulated in the literature.

It is advantageous for numerical and analytical integration to transform Eqs. (2) following the Keldysh approach [5]. To that end, we introduce the transformation $a_{\mathbf{m}} = b_{\mathbf{m}} \exp(-i \int_{-\infty}^t E_{\mathbf{m}} dt')$ ($\mathbf{m} = v, c$) together with $\mathbf{K} = \mathbf{k} - \mathbf{A}(t)$. As a result, Eqs. (2) become

$$\dot{b}_v(\mathbf{K}, t) = iF(t)d(\mathbf{K} + \mathbf{A}(t))b_v(\mathbf{K}, t) \exp(-iS(\mathbf{K}, t)) \quad (4a)$$

$$\dot{b}_c(\mathbf{K}, t) = iF(t)d^*(\mathbf{K} + \mathbf{A}(t))b_c(\mathbf{K}, t) \exp(iS(\mathbf{K}, t)), \quad (4b)$$

where

$$S(\mathbf{K}, t) = \int_{-\infty}^t \varepsilon_g(\mathbf{K} + \mathbf{A}(t')) dt' \quad (5)$$

is the classical action, and $\varepsilon_g = E_c - E_v$ is the bandgap.

A connection between Schrödinger equation and density matrix equations can be established in the following way [2]. We define band population $n_{\mathbf{m}} = |b_{\mathbf{m}}|^2$ and $\pi = b_c^* b_v$ which is related to the polarization by $\mathbf{p}(\mathbf{K}, t) = \mathbf{d}(\mathbf{K} + \mathbf{A}(t))\pi(\mathbf{K}, t) \exp(iS(\mathbf{K}, t)) + \text{c.c.}$ By multiplying Eqs. (4) with b_v^* and b_c^* and by utilizing the complex conjugate of the resulting equations, the equations for the probability amplitudes can be rewritten into differential equations for the population of valence and conduction bands and for π which are

$$\dot{\pi}(\mathbf{K}, t) = -\frac{\pi(\mathbf{K}, t)}{T_2} - i\Omega(\mathbf{K}, t) w(\mathbf{K}, t) e^{-iS(\mathbf{K}, t)} \quad (6a)$$

$$\dot{n}_{\mathbf{m}}(\mathbf{K}, t) = is_m \Omega^*(\mathbf{K}, t) \pi(\mathbf{K}, t) e^{iS(\mathbf{K}, t)} + \text{c.c.} \quad (6b)$$

where $w = n_v - n_c$ is the population difference between valence and conduction band; $s_m = -1, 1$ for valence and conduction band, respectively and $\Omega(\mathbf{K}, t) = \mathbf{F}(t) \cdot \mathbf{d}(\mathbf{K} + \mathbf{A}(t))$ is the Rabi frequency. The dephasing (T_2) term has been added phenomenologically; in principle, it could also be derived more formally by extending the theory to account for coupling to phonon bath and impurities and for electron-electron scattering [10]. If the same procedure is repeated for the original Eqs. (2), the standard two-band density matrix equations are obtained. The transformed set of equations (6) has advantages for numerical integration and for developing analytical solutions, as is demonstrated in the main text.

Determination of transition dipole moment

The interband dipole matrix element $\mathbf{d}$ can be determined by relating it to the momentum matrix element $\mathbf{p}(\mathbf{k}) = \langle u_v(\mathbf{k}) | \hat{\mathbf{p}} | u_c(\mathbf{k}) \rangle$, which was calculated by DFT and tabulated in the literature for wurtzites [6–9].

This relation can be found by starting from the identity

$$\langle \Phi_{v,\mathbf{k}} | \mathbf{x} | \Phi_{c,\mathbf{k}'} \rangle = \frac{\langle \Phi_{v,\mathbf{k}} | [\mathbf{x}, H_0] | \Phi_{c,\mathbf{k}'} \rangle}{E_v(\mathbf{k}) - E_c(\mathbf{k}')}, \quad (7)$$

where $[\cdot, \cdot]$ is the commutator. Next we use $\mathbf{p} = [\mathbf{x}, H_0]$ and approximate the Bloch wavefunction by the leading order term of the $\mathbf{k} \cdot \mathbf{p}$ method [10], $\Phi_{m,\mathbf{k}} = u_{m,\mathbf{k}=0} \exp(i\mathbf{k} \cdot \mathbf{x})$, which presents a good approximation for small $\mathbf{k}$ around the Γ -point. This yields $\mathbf{d}(\mathbf{k}) = -i\mathbf{p}(\mathbf{k} = \mathbf{0})/\varepsilon_g(\mathbf{k})$. By using the definition for the Kane parameter $E_{pj} = 2p_j^2(\mathbf{k} = \mathbf{0})$ with ($j = x, y, z$) we finally get

$$d_j(\mathbf{k}) = \sqrt{\frac{E_{p,j}}{2\varepsilon_g^2(\mathbf{k})}} \quad (8)$$

We neglect the $\mathbf{k}$ -dependence of the dipole moment, $\varepsilon_g(\mathbf{k} = 0) = 0.121$. From literature and our DFT analysis we find the Kane parameters to lie in the range $E_{px} = E_{py} = 0.248 \rightarrow 0.355$ and $E_{pz} = 0.270 \rightarrow 0.479$ [6–9] which yields dipole moments $d_x = d_y = 3.0 \rightarrow 3.37$ and $d_z = 3.12 \rightarrow 3.92$.

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